Scaling Agents for Computer Use

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Paper version: arXiv 2510.02250v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Generating multiple rollouts is straightforward, [but] reliably determining which long-horizon trajectory is correct is not.*

Wide scaling needs a **judge** that can pick the right trajectory.

Motivation: three obstacles

1. **Single runs vary.** Small errors compound across steps.
2. **Step-wise scaling commits** to one plan.
3. **Raw trajectories are dense;** most detail is irrelevant.

Evaluation setup

OSWorld · 361 tasks | WindowsAgentArena · 154 | AndroidWorld · 116 | GPT-5 / Opus 4.5 agents | Agent S3 base agent | 100-step budget | N = 10 rollouts

- **All online**; no offline accuracy.
- **Transfer:** Windows, Android zero-shot.

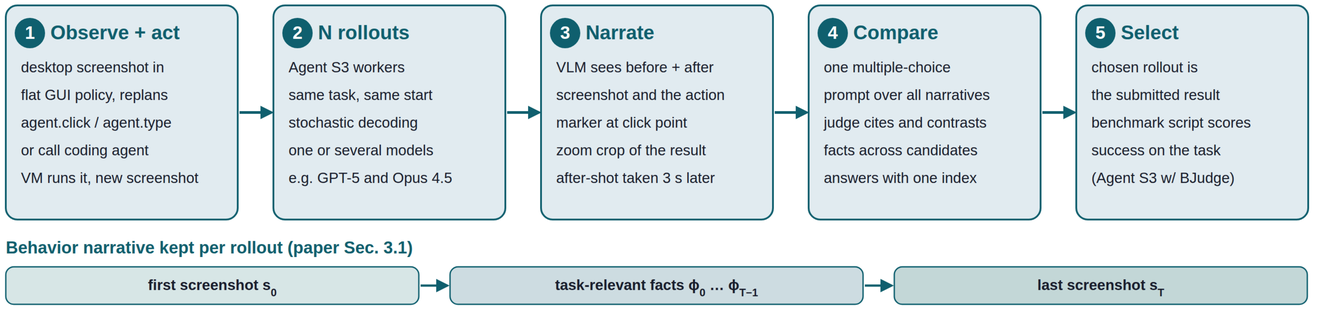
Our read (CUA Radar)

68	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; transfer to 2 benchmarks; strong baselines; equal budgets; cost; ablation; judge check.
- **Flags:** none. **Not reported:** BJudge variance; code.
- Pattern K02 (verifier rerank). Authors' numbers, not rerun.

Method: narrate each rollout, then pick one by comparison

- **Sees / acts:** desktop screenshots in; agent.click, agent.type or a coding-agent call (Python/Bash) out.
- **Narrative:** a VLM turns each before/after screenshot pair and action into facts; clicks get a marker and a zoom.
- **Signal:** one multiple-choice prompt over all narratives; the judge picks one rollout.



Our redraw (not the paper's figure): rollouts, behavior narratives, and one-shot comparative selection.

$$\hat{\tau} \in \arg\max_{\tau \in C} R(\tau, I)$$

Objective: pick the best candidate trajectory

Key results: OSWorld online success (%), 100 steps

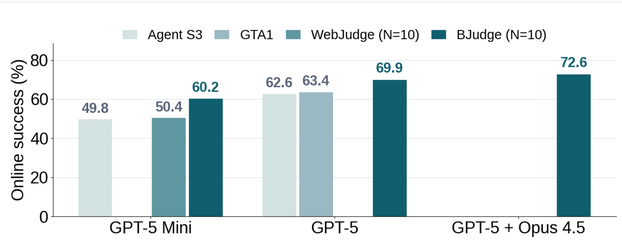


Table 1, redrawn: 361 tasks. WebJudge and BJudge use N = 10 rollouts; GTA1 is step-wise scaling.

9.2% absolute over the prior best 63.4%; human level 72.36%

SO WHAT → *Empty bar = not reported in Table 1.*

Ablation: trajectory representation (%)



Table 3, redrawn: OSWorld, 10 GPT-5 Mini rollouts.

SO WHAT → *Paper: narratives give a 3.4% improvement over the best baseline; screenshots alone, summaries and per-screenshot captions all score lower, with the same 10 GPT-5 Mini rollouts. The screenshot-only baseline passes 50/N screenshots per trajectory because of context limits. In the equal-budget check (Figure 4), WebJudge improves slightly near N=4, then plateaus and drops around N=10.*

Headline Numbers

72.6%
OSWorld 100-step success
GPT-5 + Opus 4.5; human 72.36%

- **69.9%** GPT-5, N=10 rollouts
- **71.6%** AndroidWorld, N=3
- **56.6%** WinAgentArena, N=3

Controls (all online unless noted)

CONTROL	BJUDGE	COMPARED WITH
OSWorld, GPT-5 Mini, N=10	60.2	WebJudge 50.4
Judge subset, 159 tasks	78.4% vs scripts	92.8% human-checked
WindowsAgentArena, 100 st.	56.6 (N=3)	Agent S3 50.2
AndroidWorld	71.6 (N=3)	Agent S3 68.1

Takeaway

Judge whole rollouts, not single steps: 72.6% on OSWorld, transfer on two other benchmarks.

rollout $0.72 / task | judging $0.03 / task

Scan to read the full paper

arxiv.org/abs/2510.02250v2